SYSDESIGN REFERENCE

Hot Keys and Uneven Load

A senior-level reference with tradeoffs, failure modes, and interview follow-ups for hot keys and uneven load.

Detect and diffuse disproportionate load concentrated on a small set of keys or tenants.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / HOT-KEYS

SYSDESIGN

Hot Keys and Uneven Load

CORE IDEA

Detect and diffuse disproportionate load concentrated on a small set of keys or tenants.

Reference architecture decision chain

01	02	03	04
Replication	Request Coalescing	Adaptive Partitioning	Caching

Decisions to defend

+ Replication
+ request coalescing
+ adaptive partitioning
+ caching
+ and quotas.

Failure modes to test

+ Single-shard saturation
+ synchronized misses
+ and moving the hotspot downstream.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Hot Keys and Uneven Load

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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